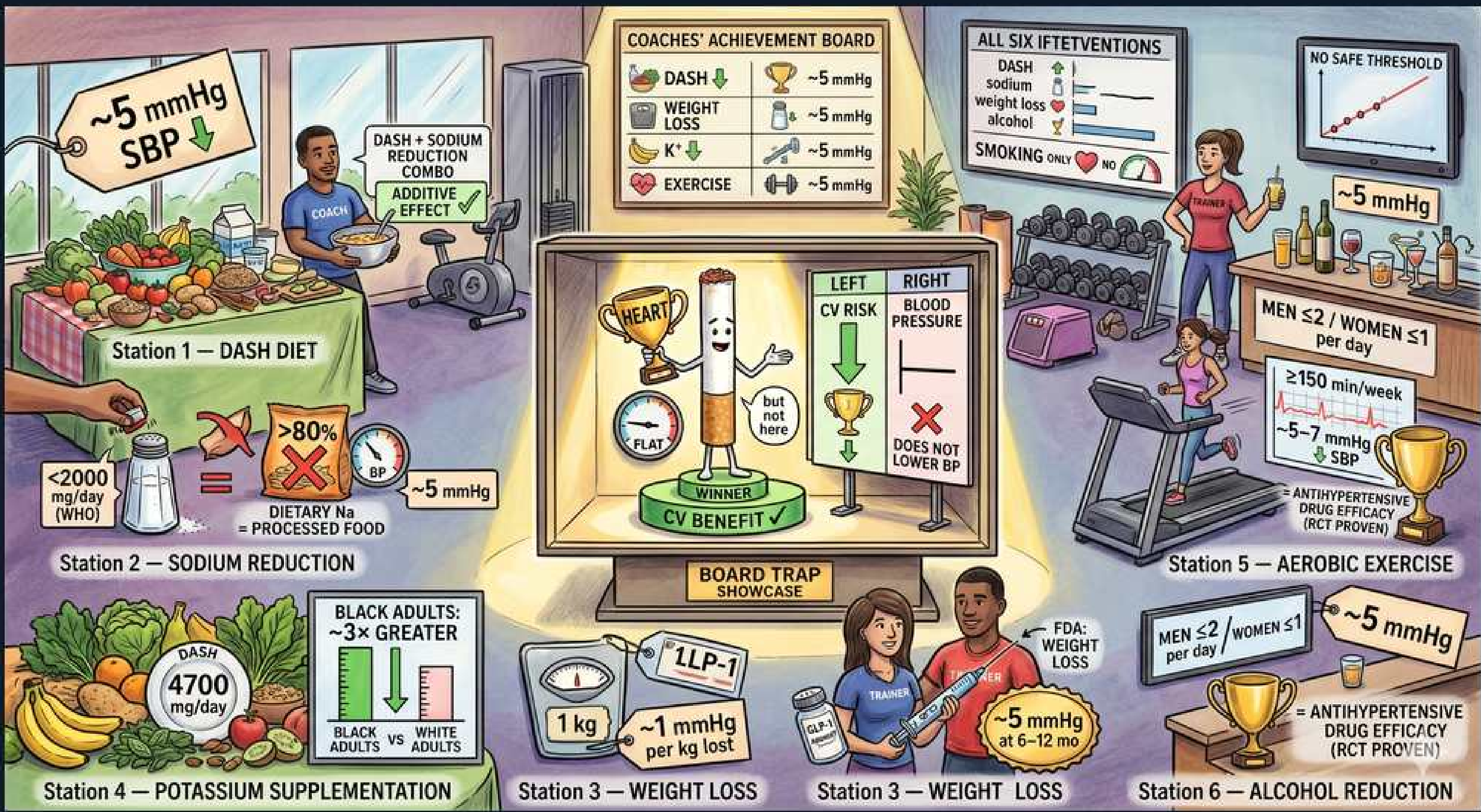
Hypertension — Lifestyle Modifications & BP Impact



1. DASH diet (~5 mmHg SBP)

WHAT YOU SEE

Station 1: DASH DIET, ~5 mmHg SBP, DASH + sodium reduction additive effect.

WHAT IT ENCODES

The DASH diet (Dietary Approaches to Stop Hypertension) emphasizes fruits, vegetables, whole grains, and low-fat dairy while limiting saturated fat, red meat, and sweets. Alone it lowers SBP roughly 5-11 mmHg, and its effect is ADDITIVE with sodium restriction (combined DASH + low-sodium can drop SBP >10-11 mmHg). It is a recommended first step for elevated BP and Stage 1 HTN at low CV risk before/with drug therapy.

BOARD TRAP

WRONG: telling a Stage 1 HTN, low-ASCVD-risk patient they need drug therapy immediately instead of a 3-6 month lifestyle trial. Discriminator — for Stage 1 (130-139/80-89) with <10% 10-year ASCVD risk, guideline-first management is lifestyle modification; drugs are added if BP remains elevated or risk is high.

2. Sodium reduction <2000 mg

WHAT YOU SEE

Station 2: SODIUM REDUCTION, <2000 mg/day (WHO), >80% from processed food, ~5 mmHg.

WHAT IT ENCODES

Limiting sodium to <2 g/day (WHO target; AHA ideal ~1.5 g/day, <2.3 g acceptable) lowers SBP about 5 mmHg. Over 70-80% of dietary sodium comes from processed/restaurant food, not the salt shaker, so label-reading and avoiding processed foods matter most. Salt sensitivity is greater in older adults, Black patients, CKD, and diabetes.

BOARD TRAP

WRONG: focusing only on removing the table salt shaker to cut sodium. Discriminator — the dominant sodium source is processed and restaurant food (>70%); counseling must target packaged/prepared foods, not just added table salt.

3. Coaches achievement board

WHAT YOU SEE

Board listing DASH, weight loss, K, exercise each ~5 mmHg.

WHAT IT ENCODES

Each individual evidence-based lifestyle measure lowers SBP by roughly 5 mmHg (DASH up to ~11), and the effects are ADDITIVE — combining several can produce drug-equivalent reductions of 10-15+ mmHg. This is why lifestyle is foundational therapy at every HTN stage, not just an alternative to medication.

BOARD TRAP

WRONG: assuming lifestyle changes give a trivial effect and skipping them once a drug is started. Discriminator — stacked lifestyle measures rival a single antihypertensive in magnitude and improve drug efficacy; they are continued alongside pharmacotherapy, not abandoned.

4. Potassium supplementation

WHAT YOU SEE

Station 4: POTASSIUM SUPPLEMENTATION, DASH 4700 mg/day, Black adults ~3x greater effect.

WHAT IT ENCODES

Higher dietary potassium (target ~3500-5000 mg/day; DASH provides ~4700 mg) blunts the pressor effect of sodium and lowers BP, with about a 3x greater effect in Black adults. Prefer potassium from food (fruits, vegetables, legumes) rather than supplements. Potassium-based salt substitutes are an effective strategy in patients without CKD.

BOARD TRAP

WRONG: prescribing potassium supplements or K-based salt substitutes to a patient with CKD or on an ACEi/ARB/MRA. Discriminator — these patients cannot excrete a potassium load and are at risk of life-threatening hyperkalemia; potassium loading is contraindicated when renal handling or renin-angiotensin blockade is in play.

5. Smoking: no BP benefit

WHAT YOU SEE

Center trophy: cigarette, CV RISK left vs BLOOD PRESSURE right; DOES NOT LOWER BP X.

WHAT IT ENCODES

Smoking cessation is the single most impactful intervention for overall cardiovascular risk, but it does NOT directly lower resting blood pressure. Nicotine causes only transient acute BP spikes; chronic essential HTN levels are not improved by quitting. The benefit of cessation is on atherosclerosis, MI, stroke, and mortality — not on the BP number.

BOARD TRAP

WRONG: listing 'smoking cessation' as the answer to 'which lifestyle change lowers this patient's blood pressure?' Discriminator — smoking cessation reduces CV RISK but not BP; the BP-lowering answers are DASH, sodium restriction, weight loss, exercise, potassium, and alcohol moderation.

6. Smoking: no safe threshold

WHAT YOU SEE

Top-right graph: SMOKING, NO SAFE THRESHOLD.

WHAT IT ENCODES

There is NO safe level of smoking — cardiovascular risk rises steeply even at low cigarette counts, and the dose-response curve is non-linear (even 1 cigarette/day carries substantial excess risk). Complete cessation, not reduction, is the goal.

BOARD TRAP

WRONG: reassuring a patient that cutting down to a few cigarettes a day makes their smoking 'safe enough.' Discriminator — the risk curve has no safe threshold and is supralinear at low doses; only complete cessation meaningfully removes the cardiovascular hazard.

7. All six interventions panel

WHAT YOU SEE

List: DASH, sodium, weight loss, alcohol... ALL SIX INTERVENTIONS.

WHAT IT ENCODES

The six evidence-based lifestyle interventions for BP are: (1) DASH diet, (2) sodium reduction, (3) weight loss, (4) aerobic exercise, (5) potassium intake, and (6) alcohol moderation. Note smoking cessation is deliberately NOT on this BP-lowering list (it lowers CV risk, not BP). All six are additive.

BOARD TRAP

WRONG: swapping smoking cessation into the list of six BP-lowering lifestyle measures. Discriminator — the canonical six are DASH/sodium/weight/exercise/potassium/alcohol; smoking cessation belongs to CV-risk reduction, a separate category.

8. Weight loss ~1 mmHg/kg

WHAT YOU SEE

Station 3: WEIGHT LOSS, 1 kg = ~1 mmHg per kg lost.

WHAT IT ENCODES

Weight loss lowers BP by approximately 1 mmHg per kilogram lost, with the greatest absolute benefit in those with the most excess weight. Even modest loss (~5-10% of body weight) yields meaningful BP reduction and improves insulin sensitivity, lipids, and OSA. Target BMI <25 kg/m^2.

BOARD TRAP

WRONG: telling an obese hypertensive that only reaching ideal body weight will help their BP. Discriminator — BP falls roughly linearly with weight (~1 mmHg/kg), so even partial, modest weight loss produces proportional benefit; perfection is not required to lower BP.

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9. GLP-1 / weight-loss drug

WHAT YOU SEE

Station 3: GLP-1, FDA weight loss, ~5 mmHg at 6-12 months.

WHAT IT ENCODES

GLP-1 receptor agonists (e.g., semaglutide, tirzepatide is a dual GIP/GLP-1) are FDA-approved for weight loss and produce a modest secondary BP reduction (~3-5 mmHg SBP) at 6-12 months, largely via weight loss. They are an adjunct in obese hypertensives, not a primary antihypertensive class.

BOARD TRAP

WRONG: starting a GLP-1 agonist as a dedicated antihypertensive in a normal-weight patient. Discriminator — the BP benefit is a downstream effect of weight loss; in patients without obesity it is not an appropriate BP therapy, and first-line antihypertensive classes should be used instead.

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10. Aerobic exercise (~5 mmHg)

WHAT YOU SEE

Station 5: AEROBIC EXERCISE, >=150 min/week, ~5 mmHg, RCT-proven.

WHAT IT ENCODES

Regular aerobic exercise — at least 150 minutes/week of moderate intensity (or 75 min vigorous), plus 2-3 sessions of dynamic/isometric resistance training — lowers SBP by about 5 mmHg, an RCT-proven effect. Isometric handgrip training also lowers BP. Benefits accrue independent of weight loss.

BOARD TRAP

WRONG: assuming exercise only helps BP if it produces weight loss. Discriminator — aerobic and resistance training lower BP through vascular and autonomic mechanisms independent of weight change; a normal-weight hypertensive still benefits ~5 mmHg from regular exercise.

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11. Alcohol reduction (~5 mmHg)

WHAT YOU SEE

Station 6: ALCOHOL REDUCTION, men <=2 / women <=1 per day, ~5 mmHg, RCT-proven.

WHAT IT ENCODES

Limiting alcohol to <=2 standard drinks/day for men and <=1/day for women lowers SBP by about 5 mmHg (RCT-proven), with the largest effect in heavy drinkers; complete abstinence gives the greatest reduction. Excess alcohol is a common, reversible cause of resistant/secondary HTN.

BOARD TRAP

WRONG: overlooking heavy alcohol use as a cause when a patient's HTN is resistant to multiple drugs. Discriminator — alcohol is a frequently missed contributor to resistant HTN and medication nonadherence; quantifying and cutting intake can restore control before escalating drug therapy.
